

ML Modelling

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- **Dynamic Time warping (DTW):** time series matching
- K nearest neighbors, one class SVM, Luminol, ExactStorm, KNNCAD
- Liquid neural nets(real time analysis)
- Bayesian Online Changepoint Detection (BOCD) algorithm, Statistical Process Control (SPC), DBSCAN
- **Spiking Neural networks:** manage huge data
- SR, SR-CNN, MCNN (Multi Scale Cnn), LSTM
- Kalman filter, GMM, HMM
- STL seasonal, trend, and residue, Isolation Forest, Classification and Regression Trees (CART), KDE (Kernal density estimation),Holt-Winters
- VMA (Vector Moving Average), VARMA (Vector Auto-Regression Moving Average), VARIMA (Vector Auto-Regressive Integrated Moving Average), and VECM (Vector Error Correction Model)- fbprophet
- STA, LTA
- Latent encoded space
- Subsequence outlier

Classification:

- DTW
- Shapelet transforms
- Spectral methods

(**Time Series Classification using AutoGluon:** convert each time series into a fixed-dimensional feature vector using libraries like tsfresh or catch22, and then use the `autogluon.tabular.TabularPredictor` with `problem_type="classification"` to classify time series.)

Anomaly Detection References:

https://conservancy.umn.edu/bitstream/handle/11299/92985/Cheboli_Deepthi_May2010.pdf?sequence=1&isAllowed=y

Autoencoders: <https://neptune.ai/blog/autoencoders-case-study-guide>

https://epub.ub.uni-muenchen.de/68482/1/BA_Park_Hyeyoung.pdf

<https://ieeexplore.ieee.org/document/7814437>

<https://proceedings.mlr.press/v71/toledano18a/toledano18a.pdf>

<http://ieeexplore.ieee.org/document/5771198>

https://www.researchgate.net/publication/265941666_Signal_Processing-based_Anomaly_Detection_Techniques_A_Comparative_Analysis

<https://d-nb.info/1220566462/34>

<https://ieeexplore.ieee.org/document/10197037>

<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10136265/>

<https://heka-ai.medium.com/anomaly-detection-in-time-series-3f36c7b4cd73>

<https://www.kaggle.com/code/victorambonati/unsupervised-anomaly-detection>

<https://turcomat.org/index.php/turkbilmat/article/download/9354/7206/16679>

Feature extraction:

<https://medium.com/@rehanmbl/extracting-time-domain-and-frequency-domain-features-from-a-signal-python-implementation-1-2-d36148c949ba>

<https://link.springer.com/article/10.1007/s10994-022-06153-4#Abs1>

https://link.springer.com/10.1007/978-3-030-61527-7_39

https://link.springer.com/10.1007/978-981-16-3637-0_4

<https://journalofbigdata.springeropen.com/articles/10.1186/s40537-020-00320-x>

Points(gen):

- QRS- T angle from ECG as cardiac indicator
- Biomedical signal analysis, book by Rangaraj M. Rangayyan
- <https://data.mendeley.com/journal/17468094>
- <https://link.springer.com/book/10.1007/978-3-540-36675-1>
- Papers on eeg, emg, ecg by topic

Hardware:

<https://www.sciencedirect.com/science/article/pii/S2452414X16000066>

<https://www.sciencedirect.com/science/article/pii/S1084804521001764>

<https://ieeexplore.ieee.org/abstract/document/7488451>

<https://link.springer.com/article/10.1007/s42452-019-1925-y>

https://link.springer.com/chapter/10.1007/978-3-030-30367-9_11

<https://ieeexplore.ieee.org/abstract/document/9355143>

https://iacis.org/iis/2018/3_iis_2018_33-41.pdf

<https://www.hindawi.com/journals/jhe/2021/6632599/>

<https://ieeexplore.ieee.org/abstract/document/8320774>

<https://ieeexplore.ieee.org/abstract/document/7474197>

https://www.researchgate.net/profile/Prajoona-Valsalan/publication/340665031_IOT_BASED_HEALTH_MONITORING_SYSTEM/links/5f60adb792851c078967a5f3/IOT-BASED-HEALTH-MONITORING-SYSTEM.pdf

<https://www.mdpi.com/1424-8220/19/9/1978>

<https://ieeexplore.ieee.org/abstract/document/7113786>

<https://ieeexplore.ieee.org/abstract/document/9034146>

<https://ieeexplore.ieee.org/abstract/document/7514585>

<https://www.diva-portal.org/smash/record.jsf?pid=diva2%3A621384&dswid=707>

<https://ieeexplore.ieee.org/abstract/document/7915565>

<https://www.mdpi.com/1424-8220/19/2/326>

<https://ieeexplore.ieee.org/abstract/document/7015961>

<https://ieeexplore.ieee.org/abstract/document/7950903>

<https://ieeexplore.ieee.org/abstract/document/7070665>

<https://www.mdpi.com/2079-6374/11/10/372>

<https://ieeexplore.ieee.org/abstract/document/8863483>

<https://ieeexplore.ieee.org/abstract/document/8124196>

<https://ieeexplore.ieee.org/abstract/document/8993839>

Streaming data analytics architecture

1. Apache Kafka and Amazon Kinesis are two popular stream processing tools
2. Combine and process the captured data to provide necessary context. This is where data integration happens; data is aggregated and transformed, usually with a streaming analytics platform or an ELT or ETL tool like Apache Spark or Hadoop.

3. You might stream processed data directly to an application or dashboards via an AWS data warehouse and query it using SQL.

A probabilistic network for the diagnosis of acute cardiopulmonary diseases:

<https://arxiv.org/pdf/1609.06864.pdf>

Disease progression models

1. Cardiovascular Diseases (CVDs)

arrhythmias, heart failure, and coronary artery disease

Abnormalities in heart rate, rhythm, and ECG waveform characteristics

2. Respiratory Diseases:

- Asthma, chronic obstructive pulmonary disease (COPD), and sleep apnea (pulse oximeters and IMU sensors)

Changes in oxygen saturation levels, respiratory rate, and movement patterns

3. Neurological Disorders:

- Parkinson's disease, epilepsy, and stroke (IMU sensors and EMG)

Changes in movement patterns, tremors, muscle activity, and gait abnormalities can provide insights into neurological health.

4. Musculoskeletal Disorders:

osteoarthritis, rheumatoid arthritis, and musculoskeletal injuries (IMU) Analysis of joint angles, movement patterns, and muscle activity

5. Metabolic Disorders:

- Diabetes and metabolic syndrome can be monitored using data from pulse oximeters, ECG sensors, and IMU sensors.

Changes in heart rate variability, blood oxygen levels, and physical activity patterns can provide insights into metabolic health and glucose regulation.

6. Infectious Diseases:

- Infectious diseases such as COVID-19 (pulse oximeters and temperature). Changes in oxygen saturation levels, respiratory rate, and body temperature

7. Stress and Mental Health Disorders:

- anxiety, depression, and stress-related disorders (ECG and IMU)

Analysis of heart rate variability, respiratory patterns, and movement patterns

8. Sleep Disorders:

Sleep disorders such as insomnia, sleep apnea, and restless legs syndrome (pulse oximeters and IMU sensors)

Patterns of oxygen saturation, movement during sleep, and sleep-wake cycles

EMG:

<https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=8050010>

<https://www.frontiersin.org/journals/neurology/articles/10.3389/fneur.2020.576729/full>

ECG

<https://nyoka.io/blog/lstm-af-detection-math-code>

IMU - Classifying activities

<https://uia.brage.unit.no/uia-xmlui/bitstream/handle/11250/2786826/Article.pdf?sequence=4>

<https://hal.science/hal-02270579/document>

<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9698430/>

PPG+ accel data for pattern recog:

<https://www.sciencedirect.com/science/article/pii/S2352340919314003>

Oximeter

PPG review+data:

<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8950880/>

<https://arxiv.org/html/2401.12783v1>

Sigpro

https://peterhcharlton.github.io/publication/ppg_sig_proc_chapter/PPG_sig_proc_Chapter_2_0210612.pdf

<https://www.nature.com/articles/sdata201876>

Python PPG toolkit:

<https://python-heart-rate-analysis-toolkit.readthedocs.io/en/latest/>